

movie buff, it would be understandable that these two movies are extremely different from each other. IMDB tends to recommend these together because of their popularity, as both are part of Top 200 Movies list. It also recommends movies based on tags allocated/stars of the movie and the like.

While all of this works, to improve it, we need to dig deeper. Most database websites like IMDB or Goodreads allow users to tag movies/books as per their understanding. For example, *2001: A Space Odyssey* (the movie) is labelled with terms like human versus computer, spaceship, spaceship setting, artificial intelligence, etc by users on IMDB (and not system-generated tags).

Such keywords are the best description of items like books/movies. These define the user's understanding of a book/movie and are much finer than genre specifications. Using natural language processing tools like WordNet, tags can be merged and can be powerful in terms of finding similar content-based items. For instance, WordNet provides relationships in between words and phrases (vertical and horizontal). It is a great tool for understanding content.

Another layer that can be added to recommendations is social network. Now, let's say my mother's birthday is coming up. The recommendation system not only reminds me of the same but also suggests items from her wishlist. Would this be compelling enough for me to make a purchase that I know my mother would love? This would help lessen the impact of the cold start problem.

Cold Start is a problem that recommendation systems suffer from. This is because when a user has lesser number of purchases/interactions, the recommendation system suggests most popular suggestions. These could either be too general or not compelling enough for the user to browse through.

By adding a social layer, or deriving data from other sources (as suggested earlier, IMDB and Goodreads are both subsidiaries of Amazon), the system can have a better knowledge about the customer (in terms of likes, dislikes, etc). None of the e-commerce ventures, however, invest in social networks (except for Amazon - Goodreads). They are largely dependent on ads on such networks.

Improvements to recommendation systems is a low-hanging fruit that would not only ensure that customers have a high repeat rate but also improve customer experience. To take an analogy from the offline world, a smart shopkeeper through his keen observation is aware which customer will make a purchase (and what item would interest him/her). This is the shopkeeper's way of generating loyalty, which is non-scalable.

This is similar for online recommendation systems (except for the scalability bit). Here, the customer displays loyalty because he/she has had a great experience on the website. The experience begins with relevant results in recommendations, which leads to purchase, thereby leading the customer to choose the platform over competition.

On a parting note, I feel recommendation systems have a lot of potential. Organisations must keep evolving their systems into smarter systems. **EFY**

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